



PROFESSIONAL SUMMARY

Software Development Engineer | 1.6 Years Experience | M.Tech CSE, IIT Kanpur. Focused on developing and deploying backend systems, AI microservices and real-time security analytics using the **Python ecosystem (FastAPI, Docker, PostgreSQL)**. Currently **developing a UEBA module** from scratch for SOC platforms, utilizing **OpenSearch** and **Bayesian behavioral modeling** to detect anomalies in live HIDS logs. Experienced in building **Generative AI** infrastructure through **RAG pipelines** and **asynchronous data workflows**. I bring a research oriented approach to solving real-world problems and building solutions from scratch. Committed to writing clean code, performance tuning, and tackling daily engineering challenges.

TECHNICAL SKILLS

- **Programming:** Python, C++, C, SQL, HTML, CSS, Javascript
- **Web & Backend:** FastAPI, Redis , ReactJS, NodeJS, Sqlite3, REST APIs, PostgreSQL, MongoDB
- **DevOps & Testing:** Docker, Microservice Architecture, Git, CI/CD Concepts , Unit Testing (Pytest)
- **AI/ML Systems:**RAG, Document Intelligence, NLP, LLM Orchestration (Langchain, LlamaIndex), Vector DBs (ChromaDB, FAISS), MLOps, Machine Learning , Scikit-learn , Numpy , Pandas
- **Cyber Security:** MITRE ATT&CK , Wireshark , Metasploit, Web Application Security, Honeypots , YARA rules , Burp Suite

PROFESSIONAL EXPERIENCE

- **Software Development Engineer | C3iHub** Jul'24 - Present
 - **UEBA Module (SOC Tool):** Developing a **User and Entity Behavior Analytics (UEBA)** feature from scratch for the SOC platform; designing end-to-end architecture for real-time behavioral anomaly detection and integration with existing security workflows.
 - **OpenSearch & HIDS Integration:** Integrated to **OpenSearch** to ingest live streaming **HIDS** (Host-based Intrusion Detection) logs; implementing anomaly detection based on failed login attempts and login-hour deviations for real-time threat visibility.
 - **Bayesian Behavioral Modeling:** Implementing **Bayesian inference** with **conjugate priors** to update user behavioral models on the go; maintaining per-user prior distributions and configurable anomaly thresholds for adaptive, learning-based detection over time.
- **Research Scholar Intern | Center for Developing Intelligent Systems(CDIS) IIT-KANPUR** Jan'24 - Mar'24
 - **Developed a retrieval-augmented generation (RAG) pipeline** in Python to answer natural language questions over a custom knowledge base (e.g., research papers).
 - Utilized sentence-transformer models for creating dense vector embeddings and implemented a **FAISS vector index** for highly efficient semantic search and retrieval.

PROJECTS

- **Policy Bot Auditor | FastAPI, Docker, ChromaDB, RAG, WebSockets | Github Link**
 - **End-to-End RAG Pipeline:** Developed a full-stack RAG microservice—from document parsing and chunking to final LLM output—using **FastAPI** and **Docker**, with **ChromaDB** synchronizing embeddings across distributed workers.
 - Real-Time Telemetry:** Engineered an asynchronous ingestion engine via **WebSockets** to stream live status updates during document indexing, ensuring a responsive and transparent user experience.
 - Forensic Retrieval & Isolation:** Automated multi-tenant data isolation and metadata mapping to provide exact source citations (page/section), reducing manual audit overhead by **80%**.
- **DeceptiSense: Threat Intelligence Pipeline(Thesis) | Python, Docker, MITRE ATT&CK**
 - **Engineered a scalable data pipeline** to capture and analyze malware artifacts from containerized honeypots, automating collection via a custom Docker-layer file extraction module.
 - **Automated threat classification** by integrating clustering algorithms with the MITRE ATT&CK framework, significantly reducing incident response and intelligence generation time.

EDUCATION

Indian Institute of Technology Kanpur	2022 – 2024
Master of Technology in Computer Science and Engineering	CPI: 7.71/10
Madan Mohan Malaviya Univ. of Technology (MMMUT)	2017 – 2021
Bachelor of Technology in Computer Science and Engineering	CPI: 7.47/10